Practice workbook · Explain how skin blood flow and sweating control body temperature

Vasodilation, vasoconstriction, shivering and sweating in hot, cold or exercise conditions, including drawn skin changes and differences in sweat volume between situations or people.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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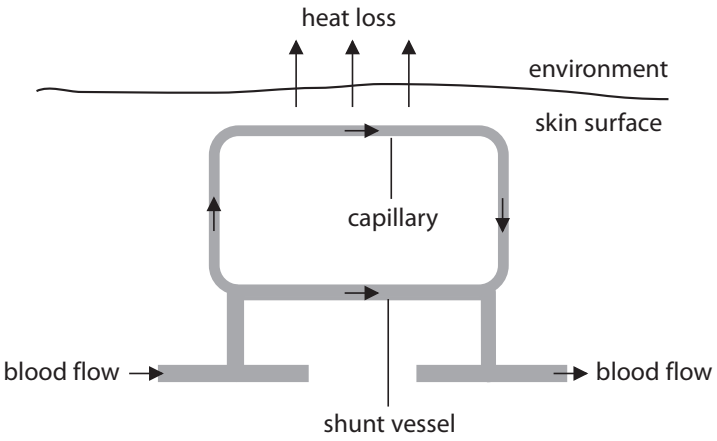
Type ID: HBI-SKIN-THERMOREGULATION-RESPONSES

1. Question 7 · part b · 4 marks

November 2020 Paper 1R

Worked-example retry · Attempt part b only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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1. Question 7 · part b · 4 marks

November 2020 Paper 1R

Worked-example retry · Attempt part b only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

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(c) Explain how the control of body temperature is an example of negative feedback.

(2)

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(Total for Question 7 = 8 marks)

1. Question 7 · part b · 4 marks · Working and answer

Working and answer

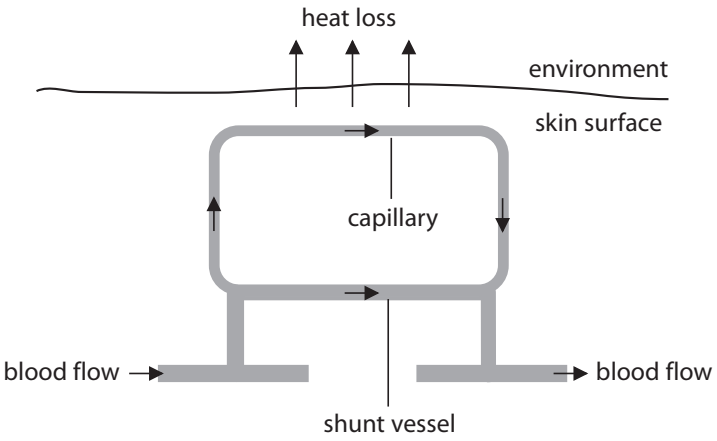
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2. Question 7 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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2. Question 7 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

(c) Explain how the control of body temperature is an example of negative feedback.

(2)

(Total for Question 7 = 8 marks)

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2. Question 7 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b · 4 marks

January 2022 Paper 2

Further practice · Attempt part b only.

3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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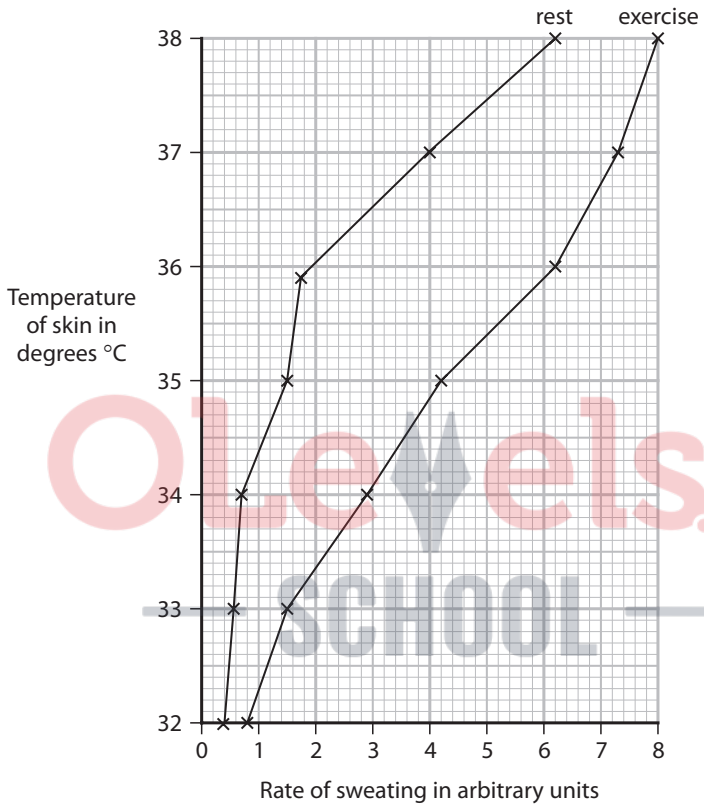
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3. Question 3 · part b · 4 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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3. Question 3 · part b · 4 marks · Working and answer

Working and answer

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4. Question 5 · part b · 3 marks

June 2022 Paper 1

Further practice · Attempt part b only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

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4. Question 5 · part b · 3 marks

June 2022 Paper 1

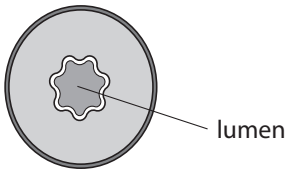
Further practice · Attempt part b only.

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(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

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(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

4. Question 5 · part b · 3 marks · Working and answer

Working and answer

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5. Question 3 · part b.ii · 4 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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5. Question 3 · part b.ii · 4 marks

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Further practice · Attempt part b.ii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

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(Total for Question 3 = 15 marks)

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5. Question 3 · part b.ii · 4 marks · Working and answer

Working and answer

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6. Question 8 · part b.i · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

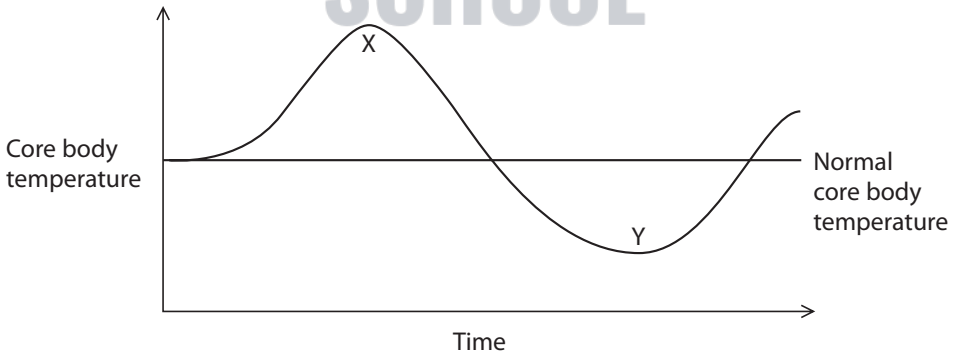
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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P 7 4 4 9 5 A 0 2 4 2 8

6. Question 8 · part b.i · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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6. Question 8 · part b.i · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

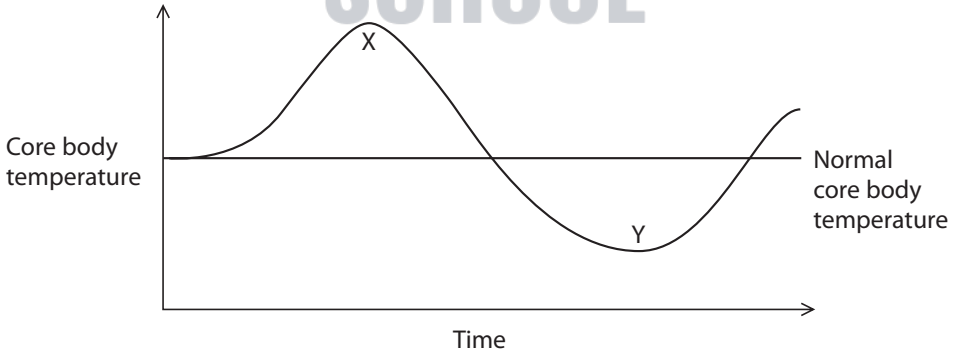
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P 7 4 4 9 5 A 0 2 4 2 8

7. Question 8 · part b.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(i) Sweating can occur at point X on the graph.

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(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

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(2)

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7. Question 8 · part b.ii · 1 mark · Working and answer

Working and answer

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8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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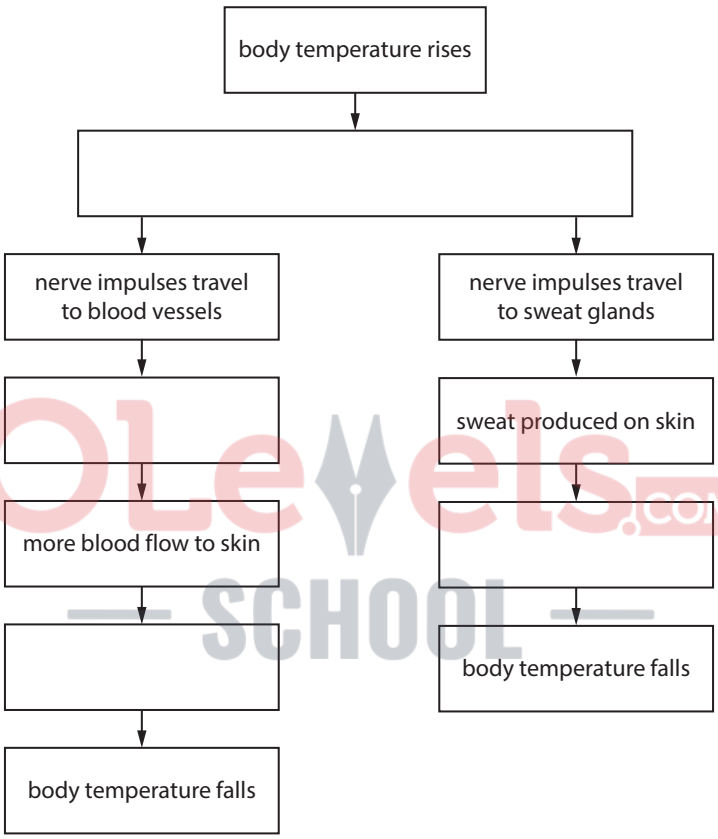
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8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

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8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

(c) Explain why it is important that body temperature should not become too high.

(4)

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(Total for Question 3 = 14 marks)

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8. Question 3 · part b · 6 marks · Working and answer

Working and answer

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9. Question 5 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

5 A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

Temperature of room in °C	Volume of sweat produced in arbitrary units	
	Sitting	Active
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

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9. Question 5 · part c · 3 marks

June 2019 Paper 1R

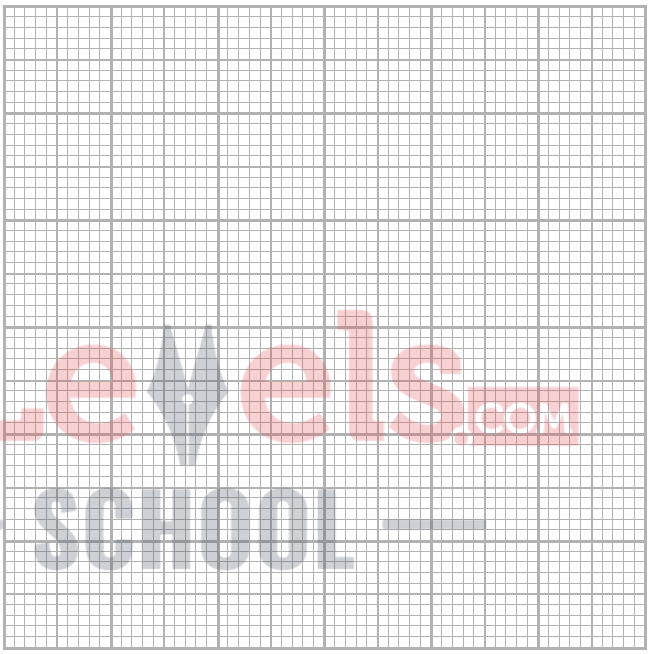
Further practice · Attempt part c only.

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(a) Plot the student's results on the grid.

Join the points with straight lines.

(5)



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(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

(2)

difference = ... arbitrary units

P 6 0 1 8 6 A 0 1 3 2 8

9. Question 5 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(c) Explain the difference between the volume of sweat produced when sitting and when active.

(3)

(Total for Question 5 = 10 marks)

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Working and answer

10. Question 3 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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10. Question 3 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

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(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

10. Question 3 · part c · 3 marks · Working and answer

Working and answer

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11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

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- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

(f) The passage states that how much we sweat varies from individual to individual and from day to day.

Explain why this is the case in people without hyperhidrosis (lines 8 and 9).

(5)

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(Total for Question 3 = 14 marks)

11. Question 3 · part f · 5 marks · Working and answer

Working and answer

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12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

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(ii) State three variables that should be controlled.

(3)

1 .....

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2 .....

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3 .....

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12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

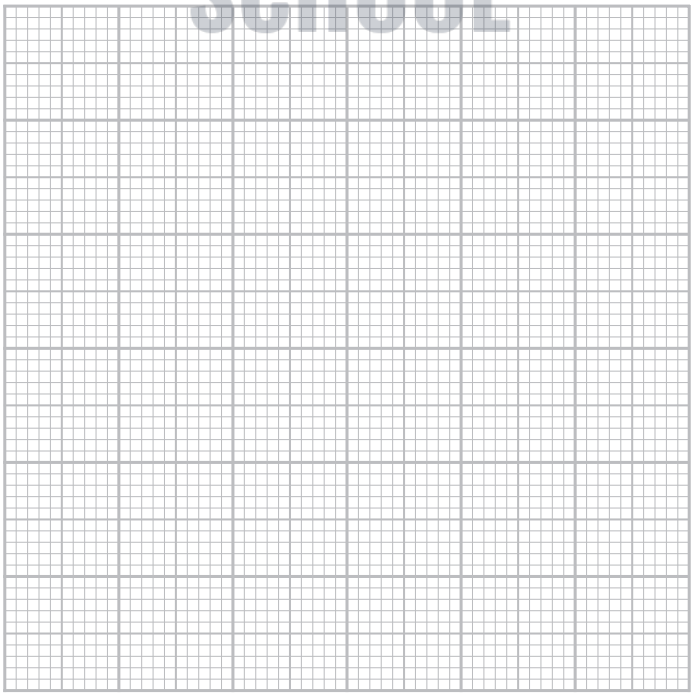
(b) The table shows the student's results.

Room temperature in °C	Amount of sweat produced by 25 cm^2 area of skin in arbitrary units	
	While inactive	During exercise
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

(i) Plot line graphs of the student's results.

Join the points with straight lines.

(6)



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P 7 2 5 8 6 A 0 1 2 2 0

12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

(ii) Explain the differences between the two sets of results.

(5)

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(Total for Question 5 = 16 marks)

12. Question 5 · part b.ii · 5 marks · Working and answer

Working and answer

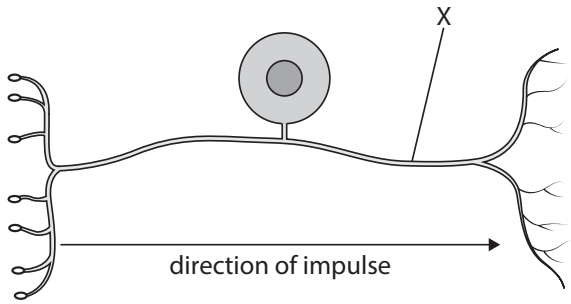
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13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

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(ii) Name the structure labelled X.

(1)

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P 7 4 6 2 6 R A 0 2 2 4

13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

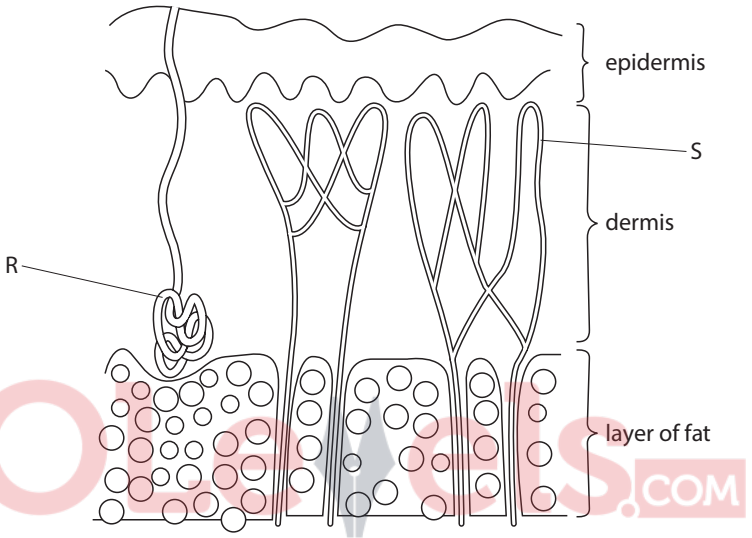
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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

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part S

.....

(ii) Describe how part R helps to cool the body.

(2)

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13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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13. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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